

Request for Proposals Assignment

Due: 2026-02-13 to
2026-02-15 @ 2200 **Weight:** 25% **Submitter:** Team

Overview

The Praxis II challenge: Effect a verified and validated sustainable improvement in the lived experience of a community in the GTA.

A request for proposals (RFP) is a document that frames a request for engineering design services. It is similar to a design brief but goes into much more detail and incorporates significantly more and better research. An RFP defines a particular project, and generally includes background information, an appropriately framed engineering design opportunity, and the objectives for any proposed design to address the opportunity.

Key stakeholders

There are four major stakeholder groups you should consider. As in all design work in Praxis, you as the authoring team are also stakeholders with interests of your own.

A specific community

As the focus of your activities in Praxis II, the community you engage with is a core stakeholder or set of stakeholders. While the specific inputs of this stakeholder will vary from community to community, at minimum, the following comprise interests of this stakeholder:

1. To be treated with respect. This includes not being overwhelmed with requests or demands by students, flooded by e-mails, texts, or telephone calls;
2. To collaborate with, not simply direct or act as a source of information for, your team; and
3. To be portrayed fairly and accurately in your work.

A community may comprise multiple groups, agencies, individuals, etc. Your team should ensure that it consults as widely as is appropriate and necessary to establish comprehensive understanding of the community.

RFP respondents (i.e. Praxis II students in Phase II of the course)

RFP respondents will be proposing both conceptual and detailed designs that address the opportunity identified in the RFP. As such, these teams require:

1. A clear engineering design opportunity framed to be appropriate for engineering design, that can be addressed by first-year EngSci students in approximately six weeks;
2. Information on and access to the community and associated stakeholders such that the responding team(s) can understand the situation from those perspectives and can develop working rapports with them; and
3. Sufficient and high quality research that provides a strong starting point for developing both an understanding of the community and background into the particular opportunity to be addressed.

The Division of Engineering Science (EngSci) and the Faculty of Applied Science and Engineering (FASE)

Both EngSci and FASE want to continue to demonstrate EngSci's leadership as “one of the most distinguished engineering programs in the world.” In addition, EngSci is interested in staying true to its vision of educating “Engineers for the World (E4TW).”¹

The Praxis II teaching team

The primary interest of the teaching team is to ensure the quality of the educational experience of all students in the course. As such, the teaching team has five main interests in the project:

1. Well-defined engineering opportunities that allow Praxis teams to generate meaningful designs;
2. Real engagement with appropriate community stakeholders that can be sustained throughout the term, both for the RFP writing team and any teams that may respond to the RFP;
3. Potential for well-designed improvements to the lived experience of community stakeholder groups– that is, quality engineering design work that clearly addresses opportunities and demonstrates students’ understanding of engineering design;
4. Well-written and well-researched requests that demonstrate an ability to gather, analyze, synthesize, and evaluate information from a range of sources and develop cogent arguments to support the core opportunity in the RFP; and,
5. Teams that are effective, are aware of their position in context, and know how their position affects their design practice.

Deliverable Components

There are three components in this assignment, and the final submission consists of two PDF files. See submission requirements for how these components should be submitted.

1. The request for proposals

The primary deliverable in response to this assignment is an RFP that frames an opportunity in a form suitable to be addressed through engineering design.

2. List of references + source extracts

The submission must include a properly formatted list of references, and all the relevant extracts from all sources used in the RFP, including interview transcripts, research paper extracts (with relevant portions highlighted), website printouts or screenshots (with relevant highlights), etc. You may choose to integrate these source extracts with your references list, or include them as a separate appendix. Submissions that fail to provide source extracts will not be assessed until they are provided.

3. Contact information for key stakeholders

For security and confidentiality, the RFP or appendices **must not** contain contact information for the community and other key stakeholders. This information **must** only be contained in a separate contact information document. Remove all identifiable community contact information (other than member name, and possible title) from the RFP and list of references. This may mean that you need to redact information from emails or meeting notes in your source extracts and appendices.

¹ The vision is explained at <https://engsci.utoronto.ca>

Assignment expectations

Requirements language in this and following sections are to be interpreted as described in RFC 2119.

Objectives

1. Demonstrate understanding of a community;
2. Develop an authentic and substantiated understanding of an engineering design opportunity that relates to the lived experience of that community;
3. Frame an engineering design opportunity using justified NGOs that can be addressed by a team of first-year EngSci students;
4. Represent your opportunity as a well-written and well-organized RFP document that communicates and interprets the stakeholders' perspectives and demonstrates research, clear thinking, and effective use of language.

Requirements

Note that violation of a constraint will **not** result in the submission being rejected (as would occur in industry). However, a penalty for violating a constraint will be levied when the assignment is assessed.

Content

1. Shall include a graphical representation of the NGOs

Format

2. Should be formatted using an 11-point serif font with 1.25-line spacing and 1-inch margins
3. Should be between 3500–5000 words in length, including tables and captions, excluding title page, abstract, table of contents, references, appendices, and non-textual elements²
4. Shall not include any identifying information of the authoring team (e.g. names, team ID, studio, etc.)
5. Shall include as part of the RFP an abstract that:
 - a. Is less than or equal to one (≤ 1) page in length, including references and non-textual elements
 - b. Includes a reference list formatted according to IEEE, APA, or CBE guidelines
6. Shall include a reference list and accompanying source extracts that:
 - a. Are formatted according to one of IEEE, APA, or CBE guidelines
 - b. Allow for easy cross-checking between citations and sources
 - c. Include source extracts for all secondary sources and notes in an appendix for all primary sources
7. Shall include a separate document with contact information for key stakeholders.
8. Shall redact contact information for key stakeholders in the RFP and appendices.

² Tables filled with text **do** count in the word constraint. Tables of words inserted as images will be treated as an attempt to circumvent the word constraint.

Submission

9. Shall submit two (2) files through Quercus
 - a file with the RFP and associated abstract, reference list, source extracts and relevant appendices
 - a file with key stakeholder contact information
10. All files **shall** be submitted in PDF format
11. Submission size (all files) **shall** be less than or equal to ($\leq$) 100 MB total for files, unless pre-cleared by the course instructors
12. **Shall** employ meaningful file names that **do not** include any identifying information of the authoring team (e.g. names, team ID, studio, etc.)

Evaluation criteria

Note that evaluation criteria are not weighted equally.

Argument and evidence

1. Credibility and quality of research in terms of balancing primary and secondary sources and corroborating information with multiple sources
2. Strength of claims, as shown by logical justification and support from evidence (i.e. contextualize sources and qualify claims; avoid unwarranted generalizations)

Framing the engineering design opportunity

3. Quality of integration of the team's position into the RFP
4. Richness, clarity, and accuracy of the description and interpretation of the community, stakeholders, and context, and of the interrelationships between these elements
5. Legitimacy of the identified opportunity

Defining stakeholder expectations and scoping

6. Appropriateness, breadth, integration and completeness of the stakeholder's perspective (need, goals and objectives)
7. Potential for a team of four to five (4–5) first-year EngSci students to design and prototype a verified and validated proposed design given the time and resources available

Organization and writing

8. Clarity, conciseness, and professionalism of the language throughout the document
9. Quality of the structure and organization to create a coherent and logical reading experience
10. Value added to the document by any non-textual content (e.g. by providing high-quality, legible visuals that are integrated with your document to enhance a reader's understanding)